

# Abduaziz Umarov

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## EDUCATION

### University of Central Florida

B.S. in Computer Science – GPA: 4.00

Orlando, FL

May 2027

## EXPERIENCE

### Software Developer

Oct 2025 – Present

#### Knight Hacks

Orlando, FL

- Contributed production features to the Forge monorepo using **Next.js** and **TypeScript**, supporting tools for a **1,000+ member** community.
- Built and optimized backend procedures and data access patterns using **tRPC** and **Drizzle**, improving average API response times by **~18%** on authenticated routes in local profiling.
- Refactored permission-gated dashboard UI to improve mobile navigation and authenticated workflows, reducing friction in member flows.
- Collaborated with student engineers via GitHub pull requests, code reviews, and CI pipelines to ship maintainable full-stack features.

### Undergraduate Research Assistant

Sep 2025 – Present

#### ISUE Lab

Orlando, FL

- Built a custom speech-error dataset by comparing **15,000+** samples of control and dysarthric recordings, labeling error types and therapy advice using **Pandas** and **NumPy**.
- Developed a therapist-oriented model using **PyTorch** to generate personalized speech feedback from error patterns, reaching **~80%** consistency with verified therapy strategies.
- Conducted pilot evaluations with dysarthric speakers to validate real-world usability and refine feedback quality with clinician input.
- Co-authored a research paper on dysarthric speech feedback systems, contributing to dataset construction and model evaluation.

## PROJECTS

### CrisisLens | TypeScript, React, Next.js, Tailwind, Databricks, Three.js, Pandas, NumPy, PyTorch

Feb 2026

- Awarded **1st place out of 234** projects in the **Databricks Geo-Insight Challenge** at **Hacklytics 2026** for building an interactive humanitarian intelligence platform with a real-time 3D globe to visualize country-level risk, funding gaps, and operational priorities.
- Built an end-to-end **Databricks** integration by connecting **Genie** to translate natural-language analyst questions into SQL-backed insights, powered by cleaned and joined data in **SQL Warehouse**.
- Engineered an ML forecasting pipeline with **PyTorch** to simulate funding scenarios and predict cross-country impacts, improving top-priority country hit rate by **~80%** and reducing false alerts in analyst queues.

### MacroMatch | TypeScript, React, Next.js, Tailwind, PostgreSQL, Prisma, NextAuth, OpenAI

Nov 2025 – Jan 2026

- Built a full-stack nutrition platform supporting **200+** items with personalized macro targets and **OpenAI**-powered meal recommendations.
- Formulated a Macro Fit Score algorithm to rank meals by proximity to protein, carb, and fat targets, reducing food selection time by **~70%**.
- Visualized daily macro intake and progress using **Chart.js**, enabling users to maintain intake within **5–10%** of target macros.

### FinBridge | TypeScript, React, Express, Node.js, Tailwind CSS, Firebase, Google Gemini, Plaid

Oct 2025

- Integrated real-time transaction ingestion via **Plaid**, normalizing balances and spending data into a structured financial schema.
- Built a **Gemini**-powered reasoning layer with structured prompts to generate context-aware financial summaries in under **500ms**.
- Designed a secure, multi-user **Firebase** data model for transaction histories, derived metrics, and LLM-ready summaries.

### FlipScript | JavaScript, React, Express, Node.js, Tailwind, OpenAI Whisper

Sep 2025

- Created a speech-to-study pipeline using **Whisper**, cutting note-taking time by **~50%** via automated transcription and summarization.
- Designed an AI-driven note structuring workflow that summarizes transcripts into concise sections while highlights key concepts for study.
- Implemented automatic flashcard generation that converts notes into editable Q&A cards optimized for active recall and spaced repetition.

## TECHNICAL SKILLS

- Languages:** C, Python, JavaScript, Java, SQL, TypeScript, HTML, CSS
- Frameworks & Libraries:** React, Express, Next.js, Three.js, Tailwind, tRPC, PyTorch, scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, OpenCV, MediaPipe
- Tools & Platforms:** Node, PostgreSQL, Firebase, SQL Server, Prisma, Drizzle, Databricks, NextAuth, AWS, Azure, Docker, Coolify, Vercel, Git, GitHub, Figma